

Representational Homomorphism Error Predicts Compositional Generalization In Language Models

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Abstract

Compositional generalization—the ability to understand novel combinations of familiar components—remains a significant challenge for neural networks despite their success in many language tasks. Current evaluation methods focus on behavioral measures that reveal *when* models fail to generalize compositionally, but provide limited insight into *why* these failures occur at the representational level. We introduce *Homomorphism Error* (HE), a structural metric that quantifies how well neural network representations preserve compositional operations by measuring deviations from approximate homomorphisms between expression spaces and their internal representations. Through controlled experiments on SCAN-style synthetic compositional tasks and small-scale Transformers, we demonstrate that HE serves as a strong predictor of out-of-distribution generalization performance, achieving $R^2 = 0.44$ correlation with OOD compositional generalization accuracy. Furthermore, our analysis reveals that model size has minimal impact on compositional structure, training data coverage exhibits threshold effects, but noise injection systematically degrades compositional representations in predictable ways. These findings provide new mechanistic insights into compositional learning and establish homomorphism error as a valuable diagnostic tool for developing more robust neural architectures and training methods.

Code and data are available at <https://github.com/ryeii/Representational-Homomorphism-for-Transformer-Language-Models>.

Keywords: Compositional Generalization, Representation, Language Model

1. Introduction

Human language understanding is characterized by systematic compositionality—the ability to combine familiar components in novel ways to understand expressions never encountered before Fodor and Pylyshyn (1988); Montague et al. (1970). For instance, once a person learns the meaning of “jump twice” and “turn” they can immediately comprehend “turn twice” without explicit instruction. This compositional capacity enables humans to generalize from limited experience to an infinite space of possible expressions.

Despite remarkable progress in natural language processing, modern neural networks struggle with systematic compositional generalization Dziri et al. (2023). Empirical studies using benchmarks like SCAN Lake and Baroni (2018), COGS Kim and Linzen (2020), and CFQ Keysers et al. (2019) have repeatedly demonstrated that while models achieve high accuracy on training distributions, they fail catastrophically when tested on novel combinations of familiar components. This limitation poses fundamental questions about whether neural architectures can truly capture the algebraic nature of human language understanding.

Current approaches to evaluating compositional generalization primarily rely on behavioral measures—comparing model outputs against expected results on held-out test sets. While such measures reveal *when* models fail to generalize, they provide limited insight into *why* these failures occur. Understanding the internal mechanisms that support or hinder compositional reasoning requires examining how models represent and manipulate compositional structure in their hidden layers, beyond surface-level performance metrics.

In this work, we introduce *Homomorphism Error* (HE), a novel structural metric that quantifies how well neural network representations preserve compositional operations. Drawing inspiration from abstract algebra, we formalize compositionality as approximate homomorphisms between expression spaces and their representations. Low homomorphism error indicates that a model’s internal representations respect compositional structure—that is, the representation of a composed expression can be systematically derived from the representations of its components. High homomorphism error suggests entangled or memorization-driven representations that fail to capture underlying compositional principles.

We evaluate our approach on a customized SCAN-style synthetic dataset that allows systematic control over compositional structure, training data coverage, and noise levels, as well as building held-out test sets to measure Out-Of-Distribution (OOD) compositional generalization accuracy. Our results show that homomorphism error successfully identifies when models learn genuinely compositional representations versus when they rely on spurious correlations or memorization strategies, as shown by comparing OOD generalization accuracies and HE measurements. This structural perspective opens new methods for developing more compositionally robust neural architectures and training procedures.

Our key contributions are:

- We formalize compositionality as approximate homomorphism between syntactic and semantic algebras, and introduce homomorphism error as a task-independent metric that assesses compositional structure directly from model representations, complementing existing behavioral evaluation methods.
- Through controlled experiments on SCAN-style synthetic compositional tasks, we demonstrate that homomorphism error serves as a reliable predictor of out-of-distribution generalization performance, achieving $R^2 = 0.44$ correlation in our noise injection studies.
- Our analysis reveals that different aspects of compositionality (unary vs. binary operations) exhibit distinct sensitivities to distributional shifts in training data, providing new understanding of compositional learning mechanisms.

2. Related Work

Compositional Generalization Benchmarks. The systematic evaluation of compositional generalization in neural networks began with Lake and Baroni’s introduction of the SCAN dataset [Lake and Baroni \(2018\)](#). SCAN demonstrated that sequence-to-sequence models, while achieving high training accuracy, failed catastrophically when tested on systematic recombinations of known components. This work established the empirical foundation for studying the systematicity challenge first articulated by Fodor and Pylyshyn [Fodor and Pylyshyn \(1988\)](#).

Building on SCAN’s foundation, subsequent benchmarks have explored different facets of compositional generalization. COGS [Kim and Linzen \(2020\)](#) introduced semantic parsing challenges with natural language, finding that Transformers achieved near-perfect in-distribution accuracy (96-99%) but much lower out-of-distribution performance (16-35%). The grounded SCAN (gSCAN) benchmark [Ruis et al. \(2020\)](#) extended compositional evaluation to situated language understanding, where meaning depends on visual context. CFQ [Keysers et al. \(2019\)](#) provided large-scale evaluation through systematically constructed train-test splits that maximize compound divergence while minimizing atom divergence.

Theoretical Frameworks. Hupkes et al. [Hupkes et al. \(2020\)](#) provided a comprehensive taxonomic framework, identifying five key aspects of compositionality: systematicity, productivity, substitutivity, localism, and overgeneralization. This theoretical foundation has guided much subsequent work in evaluating and understanding compositional behavior. Recent surveys [Sinha et al. \(2024\)](#) have further connected compositionality to broader questions of generalization and human-like reasoning in AI systems.

The field has developed several quantitative approaches to measuring compositional generalization. Keysers et al. [Keysers et al. \(2019\)](#) introduced compound divergence as a metric for assessing train-test split difficulty, finding strong negative correlations between compound divergence and model accuracy. Other work has connected systematic generalization to information entropy [Wold et al. \(2025\)](#), showing that generalization scales with the distributional properties of compositional components in training data.

Architectural Solutions. Various architectural innovations have been proposed to improve compositional generalization. Meta-learning approaches, particularly the MLC (Meta-Learning for Compositionality) framework [Lake and Baroni \(2023\)](#), have shown that neural networks can achieve human-like systematicity when optimized specifically for compositional skills. Neuro-symbolic approaches like the Compositional Program Generator [Klinger et al. \(2023\)](#) achieve perfect performance on compositional benchmarks with dramatically improved sample efficiency.

For Transformer architectures, improvements have come through auxiliary training objectives [Jiang and Bansal \(2021\)](#), curriculum learning with dataset cartography [Ince et al. \(2023\)](#), and architectural modifications such as increased depth [Merrill and Sabharwal \(2025\)](#). Graph-based semantic parsing frameworks have shown particular promise for structural generalization tasks [Petit et al. \(2023\)](#).

Internal Representation Analysis. Understanding the internal mechanisms underlying compositional behavior has been addressed through various probing methodologies [Belinkov \(2021\)](#). Work in mathematical reasoning has demonstrated that neural networks can learn compositionally structured representations that reflect sub-expression meanings [Russin et al. \(2021\)](#). Recent neuroscience-inspired work has shown evidence for compositional representations through algebraic operations on brain activity patterns [Ferrante et al. \(2025\)](#).

However, existing approaches to measuring compositionality have primarily focused on behavioral evaluation or task-specific probing. Our homomorphism error metric differs by providing a principled, architecture-agnostic measure of how well models preserve compositional structure in their internal representations, independent of surface-level task perfor-

mance. This structural approach offers new insights into the mechanistic basis of compositional generalization failures and successes.

3. Homomorphism Error as Structural Metric for Compositionality

We begin by formalizing compositionality in terms of homomorphisms between syntactic expressions and their semantic interpretations. Let $\mathcal{P}$ denote a finite set of primitives and $\circ$ a syntactic composition operator defined by a grammar G . Let $\mathcal{E}$ be the set of expressions generated from $\mathcal{P}$ using $\circ$. Each expression $e \in \mathcal{E}$ has an associated semantic interpretation $\text{JeK} \in \mathcal{S}$, where $(\mathcal{S}, \bullet)$ is a semantic algebra with composition operator $\bullet$.

Compositionality as homomorphism.

We say the mapping $\text{J-K} : \mathcal{E} \rightarrow \mathcal{S}$ is *compositional* if for all $e_1, e_2 \in \mathcal{E}$,

$$\text{Je}_1 \circ \text{e}_2 \text{K} = \text{Je}_1 \text{K} \bullet \text{Je}_2 \text{K}. \quad (1)$$

That is, the meaning of a composed expression is given by the composition of the meanings of its parts.

Approximate homomorphism in language models. Consider a language model M_θ with hidden representation function $\Phi_\ell : \mathcal{E} \rightarrow \mathbb{R}^d$ at layer ℓ . We introduce an auxiliary learnable operator $\star : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ (e.g. linear map, bilinear map, or MLP) trained to approximate compositionality at the representation level. The *Homomorphism Error (HE)* at layer ℓ is

$$\text{HE}_\ell = \mathbb{E}_{(e_1, e_2) \sim \mathcal{D}} \left[d(\Phi_\ell(e_1 \circ e_2), \Phi_\ell(e_1) \star_\ell \Phi_\ell(e_2)) \right], \quad (2)$$

where $\mathcal{D}$ is a distribution over expressions and d is a distance metric such as mean squared error (MSE). In practice, we extract pairs of compossible expressions from the training dataset where $\star_\ell$ learns to predict the representation of a composed expression from its components. We instantiate $\star_\ell$ with three operator families (linear, bilinear, MLP) and report the average error across them to avoid biasing the analysis toward a particular functional form of composition.

Interpretation. Low HE indicates that internal representations are *approximately homomorphic* with respect to the task’s compositional structure, whereas high HE suggests entangled or memorization-driven representations. Thus, HE measures the extent to which compositional structure is linearly or non-linearly decodable from hidden states, independent of task accuracy.

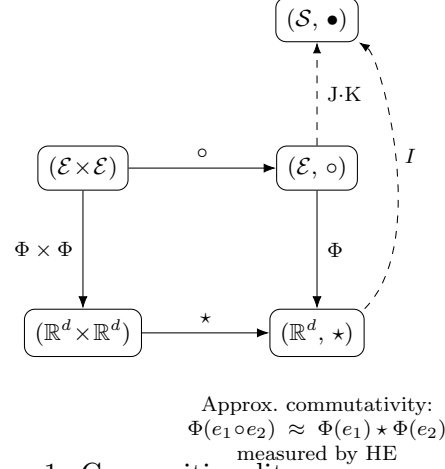


Figure 1: Compositionality as approximate homomorphism. Solid arrows form the representation-level square; dashed arrows show semantic interpretation J-K and an evaluation/readout I . Homomorphism Error (HE) quantifies how well Φ preserves composition.

4. Experiment

4.1. Dataset Construction

We design a controlled synthetic dataset inspired by SCAN-style tasks in order to probe compositional generalization. Let $\mathcal{P}$ denote a finite set of *primitives* (e.g., `walk`, `jump`, `look`, `turn`) that each map to atomic output sequences over a target alphabet Σ . We further introduce a set $\mathcal{M}$ of *modifiers* (e.g., `twice`, `thrice`) that act as unary operators on primitives, and a set $\mathcal{C}$ of *connectors* (e.g., `then`) that define binary composition.

Formally, the grammar G for input expressions is defined as

$$e ::= p \mid m(e) \mid e_1 c e_2, \text{ where } p \in \mathcal{P}, m \in \mathcal{M}, c \in \mathcal{C}.$$

The semantics $\text{Je}K$ is an output sequence in Σ^* defined compositionally by rules such as

$$\text{J}m(e)K = f_m(\text{Je}K), \quad \text{J}e_1 c e_2K = g_c(\text{J}e_1K, \text{J}e_2K),$$

where f_m and g_c are deterministic rewriting functions. For example,

$$\text{Jjump twice}K = \text{Jjump}K \text{Jjump}K, \quad \text{Jlook then walk}K = \text{Jlook}K \text{Jwalk}K.$$

To study the effect of different amount of noise in the training dataset, we introduce a finite set of *noise tokens* denoted by $\mathcal{K}$ (e.g. `foo`, `bar`, `baz`). When constructing noisy datasets, the noise tokens are inserted at random positions in the prompts, but not in the outputs. For example,

$$\underbrace{\text{foo jump bar thrice then look baz}}_{\text{input}} \mapsto \underbrace{\text{jump jump jump look}}_{\text{output}}$$

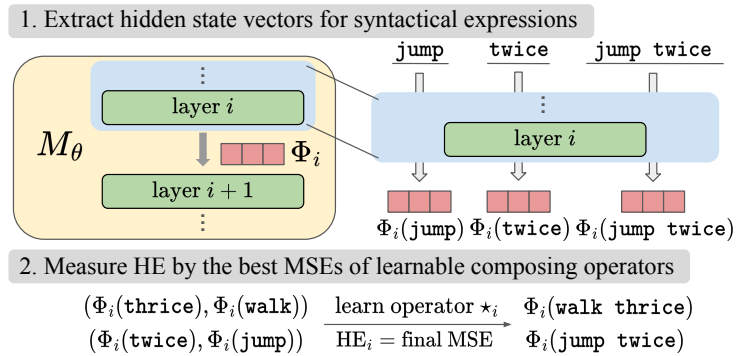
This dataset construction allows us to systematically control the number of primitives, modifiers, connectors, and noise tokens during training, and to evaluate generalization to held-out combinations.

4.2. Homomorphism Error Probe Design

Building on the general definition of Homomorphism Error (HE) in Section 3, we distinguish two forms of compositionality present specifically in the above dataset, namely *modifier HE* and *sequence HE*, by the number of parameters (unary/binary) that the compositional operation requires.

Modifier HE. Modifier homomorphism concerns unary composition $m(e)$, such as `twice` and `thrice`. For a representation function Φ at model layer ℓ , we define

$$\text{HE}_\ell^{\text{mod}} = \mathbb{E}_{(m,e) \sim \mathcal{D}_{\text{mod}}} \left[d(\Phi_\ell(m(e)), \star_\ell^m(\Phi_\ell(e))) \right], \quad (3)$$



where $\star_\ell^m : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a learned operator specific to modifier m , and d is a distance metric such as MSE. Low HE^{mod} indicates that modifiers are represented as structure-preserving transformations.

Sequence HE. Sequence homomorphism concerns binary composition $e_1 \ c \ e_2$. For representation function Φ_ℓ , we define

$$\text{HE}_\ell^{\text{seq}} = \mathbb{E}_{(e_1, c, e_2) \sim \mathcal{D}_{\text{seq}}} \left[d(\Phi_\ell(e_1 \ c \ e_2), \star_\ell^c(\Phi_\ell(e_1), \Phi_\ell(e_2))) \right], \quad (4)$$

where $\star_\ell^c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a learned binary operator associated with connector c . Low HE^{seq} indicates that connectors are represented as structure-preserving composition operators.

When calculating Modifier HE, we extract (primitive, modifier, combined) triples from the training dataset, where the combined representation is the mean pooling of consecutive primitive and modifier token representations. For Sequence HE, we extract (part1, part2, combined) triples where parts are primitive-initiated segments and the combined representation is their average. Noise tokens are not compossible with any other tokens and are not included for HE calculation, thus only the HEs of meaningful compositions are measured.

4.3. Experiment Designs

We conduct three families of controlled experiments to investigate how model architecture, training data composition, and noise affect both out-of-distribution (OOD) generalization and internal compositional structure as measured by the homomorphism errors. All experiments used a fixed OOD test sets composed of held-out expressions containing 5 to 12 primitives. 200 unique expressions were sampled from the space of primitives with each number of primitives. Layer-wise HE^{mod} and HE^{seq} are computed to determine whether deeper models learn more compositional internal representations.

Model architecture and training. All models are decoder-only transformers trained with a causal language modeling objective. All experiments used a set of fixed hyperparameters: hidden dimension $d = 128$, number of attention heads $h = 4$, feedforward dimension 256. Inputs are tokenized and passed through learned embeddings with positional encodings. The final hidden states are projected to the vocabulary space via a linear output layer. Models are trained using cross-entropy loss with teacher forcing, optimized with Adam ($\beta_1 = 0.9, \beta_2 = 0.98$), learning rate 10^{-4} , and batch size 64. Training runs for 50 epochs with early stopping on validation loss. All experiments share the same optimization settings to isolate the effects of model depth, training sparsity, and noise.

1. Model size ablation. We vary the number of transformer layers $L \in \{1, 2, \dots, 10\}$. For each configuration, we train models on a fixed dataset containing 2 primitives with no noise.

2. Training data sparsity. To probe the effect of training data coverage, we construct datasets with the expressions containing increasing numbers of primitives: 1, 2, 3, 4, always keeping num_noise = 0. For each sparsity level, models are trained with a fixed architecture ($L = 4$ layers) and evaluated on the same OOD test sets as above. This experiment tests whether reduced training coverage leads to higher homomorphism error and worse generalization.

3. Noise injection. We evaluate the robustness of learned compositional representations to spurious tokens by constructing training datasets with 2 primitives and varying

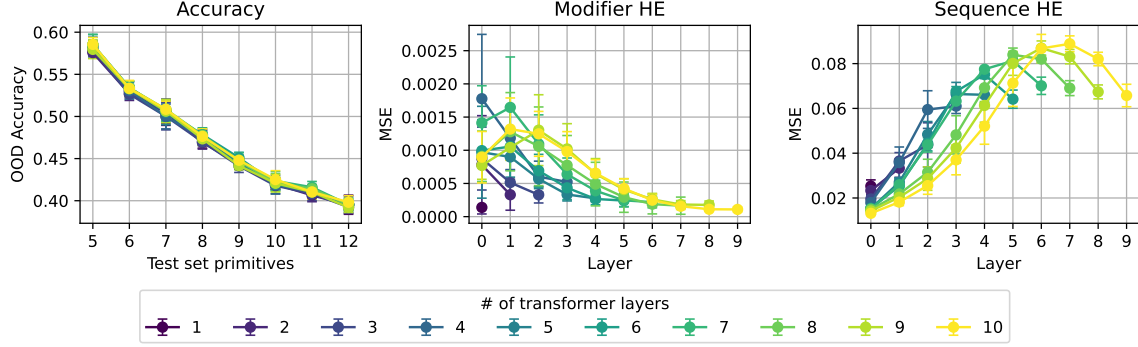


Figure 3: Model size ablation results. Lines represent number of transformer layers $L \in \{1, 2, \dots, 10\}$ in the language model.

numbers of randomly inserted noise tokens $\text{num_noise} \in \{0, 1, \dots, 15\}$. Models are trained with 4 layers and evaluated on OOD sequences without noise. Both accuracy and homomorphism error are tracked to determine whether noise disrupts compositional representations.

Evaluation metrics. For each experiment, we report:

- **OOD accuracy:** average fraction of correctly predicted output sequences on held-out test sets.
- **Modifier HE:** layer-wise MSE between representations of $m(e)$ and $\star_\ell^m(\Phi_\ell(e))$ across all modifiers in the dataset. Values are final MSEs averaged across linear, bilinear, and MLP operators.
- **Sequence HE:** layer-wise MSE between representations of $e_1 c e_2$ and $\star_\ell^c(\Phi_\ell(e_1), \Phi_\ell(e_2))$ across all connectors. Values are final MSEs averaged across linear, bilinear, and MLP operators.

Seed averaging and error bars. Each experiment is repeated with 5 random seeds to control for stochasticity in dataset generation, model initialization, and training. Reported metrics include mean and standard deviation across seeds, which are visualized as error bars in all plots.

Computing Resources. All experiments (including 5 random seeds per configuration across all ablations) were run on a single Apple M1 chip with 8GB of memory. The full experiment suite completed in approximately 30 minutes.

4.4. Experiment Results

Our experiments reveal three key findings about the relationship between model architecture, training data composition, compositional structure, and out-of-distribution generalization.

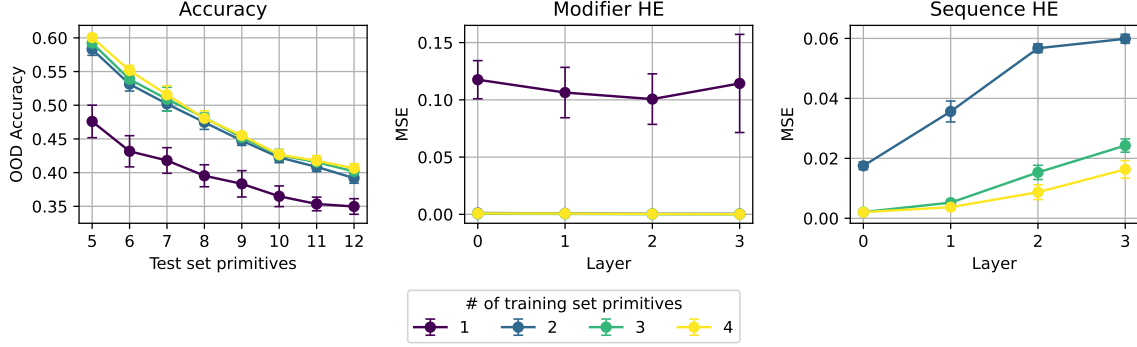


Figure 4: Training data sparsity results. Lines represent number of primitives in the expressions that the training set contains up-to. 4 meaning the training set contains expressions with up-to 4 primitives.

Model size has minimal impact on compositional generalization. Figure 3 demonstrates that increasing model depth from 1 to 10 transformer layers yields negligible improvements in OOD accuracy, with all configurations achieving approximately 40–60% accuracy across different test complexities. More importantly, both modifier and sequence homomorphism errors remain remarkably stable across model sizes, with variations on the order of 10^{-3} . The modifier HE shows slight fluctuations between layers but no systematic trend, while sequence HE exhibits similarly minimal variation. This suggests that for our controlled compositional task, representational capacity beyond a single layer provides little benefit for learning compositional structure, and that the fundamental challenge lies not in model expressivity but in the compositional inductive biases encoded during training.

Training data coverage exhibits threshold effects on compositional learning. The training data sparsity results in Figure 4 reveal a sharp threshold effect in compositional generalization. Models trained with only 1 primitive achieve significantly lower OOD accuracy (35–47%) and substantially higher modifier HE (0.015–0.018 MSE) compared to models trained with 2 or more primitives (40–60% accuracy, 0.005–0.007 MSE modifier HE). This dramatic performance gap occurs because training sets with a single primitive cannot cover binary connectors, leaving models unable to learn compositional rules for sequence construction. However, once the training set includes sufficient coverage to span the full compositional domain (2+ primitives), the marginal benefit of additional primitives rapidly diminishes. Models trained with 2, 3, or 4 primitives show nearly identical OOD accuracy and homomorphism error profiles across layers, indicating that compositional generalization depends primarily on *structural coverage* rather than *data volume*.

Noise injection systematically degrades compositional representations. The most striking results emerge from the noise injection experiment (Figure 5). Unlike model size and training data sparsity, the number of randomly inserted noise tokens exhibits a strong, monotonic relationship with both generalization performance and internal compositional structure. Models trained with increasing noise levels (0–15 tokens) show consistently de-

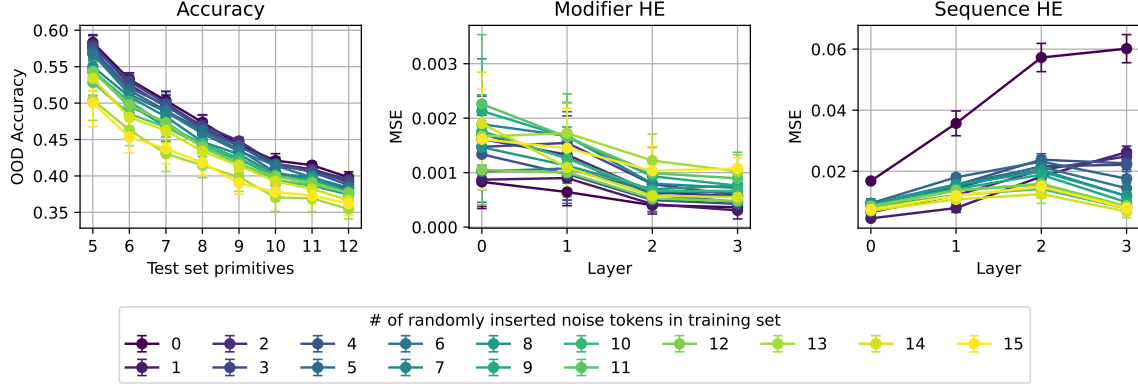


Figure 5: Noise injection results. Lines represent number of noise tokens inserted in training data.

graded mean OOD accuracy, declining from approximately 47% with no noise to 42% with 15 noise tokens. This degradation is accompanied by a systematic increase in modifier HE from approximately 0.002 MSE to 0.012 MSE, while sequence HE remains relatively stable across noise levels.

The predictive power of our homomorphism error metric is most clearly demonstrated in Figure 6, which plots the relationship between mean modifier HE and mean OOD accuracy across all noise conditions. Polynomial regression analysis reveals a highly reliable relationship, with $R^2 = 0.44$ for linear and quadratic fit. This strong correlation indicates that modifier HE serves as an effective predictor of out-of-distribution compositional generalization, capturing the degree to which models represent modifiers as structure-preserving transformations rather than memorized input-output mappings.

These results collectively suggest that compositional generalization in language models depends less on raw model capacity or training set size, more on the *structural integrity* of learned representations. Noise injection appears to interfere specifically with the model’s ability to learn modifier operations as compositional functions, while preserving sequence-level compositional structure. This dissociation provides evidence

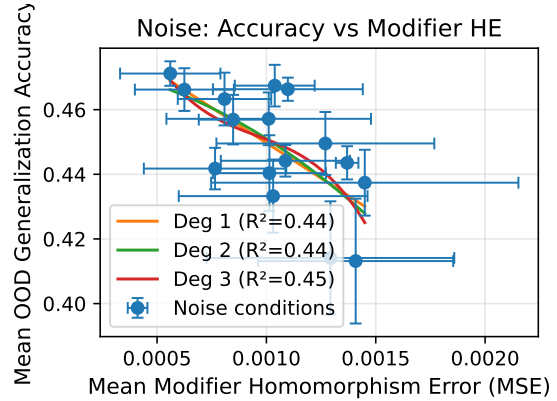


Figure 6: Analysis result of the noise injection experiment. Correlation between mean OOD generalization accuracy and mean modifier HE is shown. Polynomial regression with various degrees is conducted and R^2 is reported.

that different aspects of compositionality (unary vs. binary operations) may be learned through distinct mechanisms and exhibit different sensitivities to distributional shifts in training data.

5. Discussion and Future Work

Our results provide several key insights into the mechanisms underlying compositional generalization in neural networks. The strong predictive relationship between homomorphism error and out-of-distribution performance ($R^2 = 0.44$) suggests that compositional failures are fundamentally rooted in representational structure rather than surface-level pattern matching. This finding supports theories that emphasize the importance of algebraic structure in neural representations [Fodor and Pylyshyn \(1988\)](#) and provides empirical evidence for the homomorphism perspective on compositionality.

The dissociation between modifier and sequence homomorphism errors reveals that different aspects of compositional structure are learned through distinct mechanisms. Our noise injection experiments show that spurious tokens primarily disrupt the learning of modifier operations (unary functions) while leaving sequence composition (binary operations) relatively intact. This suggests that unary and binary compositional operations may rely on different neural circuits or learning dynamics, opening new avenues for targeted architectural interventions.

Interestingly, our finding that model depth has minimal impact on compositional generalization challenges common assumptions about the relationship between representational capacity and systematic generalization. Instead, our results point to the quality of compositional structure in representations as the critical factor, rather than raw model expressivity.

5.1. Limitations

Several limitations constrain the generalizability of our current findings. First, our experiments are conducted on synthetic data with precisely controlled compositional structure. While this enables rigorous analysis of the homomorphism error metric, real-world language presents additional complexities including semantic ambiguity, context dependence, and irregular constructions that may not conform to strict compositional principles.

Second, our evaluation focuses on relatively small Transformer models with controlled architectures and hyperparameters. The behavior of homomorphism error in large-scale pretrained language models remains an open question, particularly given evidence that scale can partially overcome compositional limitations [Brown et al. \(2020\)](#).

Third, our current framework assumes discrete, well-defined compositional operations. Extending the homomorphism error framework to capture more nuanced forms of compositionality—such as semantic composition in natural language where meaning is not strictly algebraic—presents both theoretical and computational challenges.

5.2. Future Directions

A natural next step is applying homomorphism error analysis to established compositional benchmarks using natural language, including SCAN, COGS, and CFQ. This would validate

the metric’s utility beyond synthetic settings and potentially reveal why certain architectural innovations succeed where others fail. Investigating how homomorphism error scales with model size and pretraining data could provide insights into whether the compositional capabilities of large language models emerge from improved representational structure or alternative mechanisms.

Furthermore, our predictive framework opens possibilities for compositionally-aware architecture search and training procedures. Future work could explore using homomorphism error as an optimization objective or regularization term, directly encouraging models to learn structured representations during training rather than discovering compositional failures post-hoc.

6. Conclusion

We introduced homomorphism error as a structural metric that formalizes compositionality as approximate homomorphisms between expression spaces and their representations. Our results demonstrate that homomorphism error reliably predicts out-of-distribution performance ($R^2 = 0.44$), revealing that representational structure, not surface-level pattern matching, underlies compositional failures. The dissociation between modifier and sequence homomorphism errors shows that different compositional aspects are learned through distinct mechanisms, with unary operations particularly vulnerable to spurious correlations. Crucially, our findings show that the quality of learned structure—rather than raw model capacity or architecture—determines systematic generalization capability. This framework provides both theoretical insights into the systematicity debate and practical diagnostic tools for developing more compositionally robust neural architectures.

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